

Effect of the COVID-19 pandemic on crowding aversion in public transport and transport mode choice: The case of Santiago, Chile

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ABSTRACT

Mode choice modeling has shown a significant crowding effect over travel time perception. But the recent COVID-19 pandemic and the related sanitary measures have changed perceptions of overcrowding on the mode choice. This study aims to quantify these changes through a hybrid discrete choice model using data collected through a revealed preferences survey applied in two time periods in Santiago, Chile. Although the overcrowding aversion was included in the choice model as a latent variable, that had a significant effect on the mode choice of people only for the set of responses belonging to the first time period, which implies a reduction of the initial pandemic impact. Results also show that women and people younger than 35 years old have higher levels of crowding aversion. By analyzing the disutility generated by each alternative mode, it is determined that the bicycle is preferred over the car for trips of distances less than 4 km. In the same way, for trips of less than 3 km, the walk has a preference over public transport. Regarding the marginal rates of substitution between travel and walking time in public transport, for trips of 30 min with an overcrowding level of 6 pax/m², women over 35 years are willing to trade 2 min of travel for 1 min of walking. When comparing the rates obtained in a pre-pandemic study, it is obtained that for trips of less than 9 min, people in the pandemic have more preferences for walking over spending time inside public transport.

1. Introduction

Studying the main elements that affect mode choice election is essential to operate and improve any transport system. Travelers' choices depend on different factors, such as travel time, waiting time, walking time, or cost, as well as perceptions such as crowding aversion, which relates to the refusal of traveling on highly congested vehicles. This last element has been studied by different authors that show that the crowding levels have a significant effect on travel time perception (Basu and Ferreira, 2021; Haywood et al., 2017; Kroes et al., 2014; Li and Hensher, 2011; Hörcher et al., 2017; Márquez and Poveda, 2019). Traveling in low crowding levels results on a lower disutility than traveling in high crowding levels due to factors such as low security, discomfort, harassment, etc. Therefore, travel time by self does not give the complete information about a disutility generate by spending time on public transport modes, such as metro or bus, which can experience higher crowding levels than other modes.

This study focuses on analyzing mode choice in Santiago, Chile, with an emphasis on public transport's crowding. Crowding aversion in

Santiago's public transport system has been studied in the literature. Batarce et al. (2015) showed a no-linear effect of crowding over travel time. The authors concluded that travel with a high crowding level has a double disutility than travel with a low crowding level. Tirachini et al. (2017) concluded that metro users are more willing to pay to reduce travel time when the crowding level is high. Soza-Parra et al. (2019) showed that people younger than 35 years old are the most sensitive about travel with high crowding levels. While these studies provide relevant insights, all of them were applied before the COVID-19 pandemic.

The COVID-19 pandemic has changed the landscape in terms of mobility patterns worldwide, adding a social distance aspect due to sanitary reasons (e.g. 1 or 2 m of distance between people) and a potential contagion fear (WHO, 2020). Consequently, it is reasonable to consider a change in perceptions since crowding now has new dimensions. Traditionally, the connotation of the crowding effect has been related with comfort. Nonetheless, some works show other dimensions around the crowding effect like stress (Cantwell et al., 2009). Now we can think that healthiness and the necessity to avoid COVID-19

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transmission is a new (and maybe more relevant) dimension of crowding aversion (Mashrur et al., 2022; Parker et al., 2021; Zafri et al., 2022).

The health situation changed activities and mobility due to politics applied to reduce virus propagation also had the objective to reduce the mobility around the city, as quarantines and home-office. Therefore, public transport declined by 80% in European countries at the early stages of the pandemic (Tirachini and Cats, 2020); likewise, in Santiago de Chile, the use reduction was about 55% for buses and 45% for the metro (Astroza et al., 2020). The long-term effects of the pandemic are still to be seen. While ridership has increased worldwide, public transport usage is expected to decrease in comparison to the pre-pandemic situation (Currie et al., 2021).

It is important to consider that public transport is perceived as the most dangerous mode, which has generated a massive user migration from this mode to others after quarantines finish (Tirachini and Cats, 2020; Zhang et al., 2021). This negative image relates to high crowding levels during peak hours and ventilation in public areas, and this last factor is essential to reduce the spread of COVID-19 (Sun and Zhai, 2020). Gkiotsalitis and Cats (2021)

Identify intervention measures that can support public transport service providers in planning and operating in a post-pandemic context, such as stop-skipping services and timetabling.

Additionally, we must consider that if we do not incorporate the crowding effect in transport modeling, we could get over or sub-estimations of public transport demand (Batarce et al., 2016; Tirachini et al., 2013). In the same line, Cats et al. (2016) and Pel et al. (2014) show that it is essential to incorporate this effect to evaluate an infrastructure project. Now, given the changes brought about by the pandemic is necessary to know if we must adjust demand estimations because users have migrated to the car from public transport are risky to their sustainability (Basu and Ferreira, 2021).

The COVID-19 pandemic has changed mode shares in different ways. On one hand, public transport has been the most affected mode, reducing its attractiveness and therefore its ridership (Cho and Park, 2021; Esmailpour et al., 2022). Hu and Chen (2021) show that this reduction is not even across different segments of the population, as socioeconomic disparities exist due to lower income people being more captive users of public transport. While the main shifts have tended to be towards private modes, such as cars and taxis (Das et al., 2021; Loa et al., 2022; Melo, 2022), which has serious consequences in terms of urban sustainability, COVID-19 has also affected active modes that act as feeders to public transport, such as shared bikes (Kim and Cho, 2022).

Thus, it is required to estimate the actual effect of the pandemic and we also need to know if the users will maintain these changes and the negative perception of public transport. This way, this study analyses possible changes in perceptions in the pandemic scenario and proposes public policies and action plans to the planning entities. Therefore, we look to analyze if people have changed their perception about travel time and crowding level to choose a travel mode. We hypothesize that the pandemic has been a significant effect on crowding perception. Still, this initial effect decreased over the months because Chile had a successful vaccine process, and the general knowledge about the pandemic has increased. Our methodology is based on an applied revealed preference survey and used this data to estimate a hybrid discrete choice model. In the model, we included a latent variable about crowding aversion as a systematic taste variation on the travel time of public transport.

The remainder of this paper is structured as follows. The second section describes the modeling framework and data collection process. The third section presents and analyses the modeling results. Finally, the fourth section presents a discussion and the conclusions of the study.

2. Methodology and data collection

This section first describes the hybrid discrete choice modeling framework used in the study. It then presents the survey design and

crowding perception indicators used to estimate latent variable. Finally, it describes how the level of service information was obtained.

2.1. Modeling framework

The study's methodology is based on a hybrid discrete choice model, which considers both tangible and latent variables to understand mode choices. We follow a Random Utility Theory (McFadden, 1974) to understand mode choices, through a Multinomial Logit model (Williams and Ortúzar, 1982). The most common approach to understand mode choice is to use tangible attributes k such as travel time and cost in the utility function (Equation (1)) of each alternative i for an individual q . This least to explanatory variables X_{ik} with parameters β_{ik} to be estimated.

$$U_{iq} = \sum_k \beta_{ik} \cdot X_{ik} + \varepsilon_{iq} \quad (1)$$

But also, non-tangible attributes can provide relevant information and help in explaining people's choice. These non-tangible variables are called latent variables, and they are estimated by a Multiple Indicator Multiple causes (MIMIC) model. The MIMIC has two equations, one of them is the structural equation (Equation (2)) that describes the value of the latent variable η_{ilq} is generally defined using characteristics r from individuals. The other is the measurement equation (Equations (3) and (4)), where the latent variable l explains the value of y_{ipq} for perception indicators p . α_{ilr} and γ_{ilp} are the parameters to estimate together with errors v_{ilq} and ξ_{ipq} .

$$\eta_{ilq} = \sum_r \alpha_{ilr} \cdot S_{ilr} + v_{ilq} \quad (2)$$

$$z_{ipq} = \sum_l \gamma_{ilp} \cdot \eta_{ilq} + \xi_{ipq} \quad (3)$$

$$y_{ipq} = \begin{cases} Y_1 & \text{if } \tau_{1ip} \leq z_{ipq} \\ Y_2 & \text{if } \tau_{1ip} < z_{ipq} \leq \tau_{2ip} \\ \vdots & \\ Y_N & \text{if } \tau_{(N-1)ip} < z_{ipq} \end{cases} \quad (4)$$

When the indicators have an ordered semantic scale, the measurement equation involves using an Ordered Logit model (McKelvey and Zavoina, 1975). Parameters τ_{nip} correspond to the thresholds of the scale. In this study, we used a grade scale from 1 to 7 (the grading system in Chile), and we assumed symmetry on the thresholds around grade 4 (which is the passing grade in Chile). This means, for example, that the dissatisfaction associated with grade 1 has the same magnitude that satisfaction associated with grade 7. Therefore, we estimated only three differences of thresholds around grade 4, denoted by δ_{lp}^k .

Finally, Equation (5) presents the resulting utility function considering a latent variable with parameter θ_{il} . The hybrid discrete choice model, consisting of Equation 2 through 5, is estimated simultaneously (Raveau et al., 2010).

$$U_{iq} = \sum_k \beta_{ik} \cdot X_{ik} + \theta_{il} \cdot \eta_{ilq} + \varepsilon_{iq} \quad (5)$$

2.2. Survey design

To understand and model mode choices in the COVID-19 context, we designed a revealed preferences survey. It was important for the purposes of the study to have real data, instead of stated preferences, to draw relevant and reliable conclusions from the results. The survey was brief and had three groups of questions; the first question group characterized the most common trip people had done in the previous week, gathering information about the mode chosen and other trip characteristics, such as time of day, origin and destination. By asking about the most common trip, we are mostly focusing on commuting trips. While

commuters constitute a large share of all weekly trips, and therefore would make sense to focus on them, less frequent travel purposes could be left outside the scope of the study. The second question group was about crowding perception in public transport, obtaining three crowding aversion indicators to be used in the MIMIC model. The last question group collected sociodemographic information of each respondent, such as gender, age, income, car and bicycle availability, and number of people with risky health conditions in the household.

The three crowding aversion indicators are described in Table 2–1. It is important to consider that they were presented under a pandemic context. Respondents had to provide a grade from 1 to 7 according to how represented they felt by each statement, considering 1 as “completely disagree” and 7 as “completely agree”. It can be seen that the grades reported in both time periods tend to have very similar distributions, and therefore the travelers behavior in terms of the attitudes captured by the indicators do not vary significantly between both time periods.

One of the main objectives of the study was to determine variation between two time periods in the pandemic and analyze if the initial effect of the pandemic over mode choice changed over time or not. Thus, the survey was applied in two time periods: November 2020 as lockdowns in Chile started to ease and April 2021 during the initial stage of a second wave of cases. None of the periods considered lockdowns, so people had freedom of traveling. Also, during the first period nobody in Chile had been vaccinated, while during the second period, around 50% of people had vaccinated at least with one dose. The vaccination process could influence how people perceive the virus and their contagion risk.

2.3. Level of service estimation

Since we applied a revealed preferences survey, we had to estimate the level of service for each alternative mode, obtaining the in-vehicle travel time, waiting time, and walking time. Each of these elements was computed using the Application Programming Interface (API) of Google Maps Platform. Then we could assign to each person their respective service level considering the origin, destination, and time of day reported in the survey. This information could be obtained for cars, public transport, and walking as the API only included these alternatives for Santiago. Therefore, in the case of the taxi, we considered the same travel time than car and a constant waiting time of 5 min. For bicycles, we took the walking's distance and assumed a speed of 8 km/h to estimate the travel time.

It is important to consider that the API's public transport waiting time assumes compliance with a timetable, which means assuming regular waiting time (a waiting time factor $\tau = 0.5$). But while metro in Santiago is extremely regular, this assumption does not hold for buses, whose headways have empirically shown to follow an Exponential distribution ($\tau = 1$). This way, the waiting time for the buses were adjusted to better represent the travel experience, according to Equation (6), considering the effect of common lines (Chriqui and Robillard, 1975), where a traveler can board the first bus line from a set, instead of waiting for a specific line.

$$\overline{TE} = \frac{\tau}{\sum_{i \in S} f_i} \quad (6)$$

The public transport cost corresponds to the official fares provided by the managing agency, DTPM (2021), which depend on the time of

day whether the trip uses metro or not. For the taxi, we used the actual fare of Cabify in Santiago, a basic rate of 0.36 USD plus 0.15 USD for every 60 s of travel. For the car, we used average performance about gasoline consumption of 15.5 Km/L (EPA, 2021) for one of the most seller cars (Hyundai Accent). We decided not to include highway toll because we know if the respondent often uses it. For the bicycle, we used an average monthly maintenance cost of 5.95 USD, and then we split this cost for the number of kilometers that a bike user does if daily go on for 8 km, allowing us to have a cost per kilometer.

Finally, we needed information on the crowding levels for public transport modes. In the case of buses, we had the density (pax/m²) for each service and period provided by the planning authorities of buses (DTPM). The densities were assigned to the travel itineraries from the API. In the same way, Metro de Santiago provided density information for each metro segment in the network every 15 min. Therefore, for metro we have more accurate information than for buses. As a result, we obtain densities for public transport as a whole. These densities are slightly higher during the 1st time period (mean of 2.91 pax/m² and standard deviation of 2.07 pax/m²) than in the 2nd time period (mean of 2.45 pax/m² and standard deviation of 1.62 pax/m²). On both time periods we have trips that cover a wide spectrum of densities (10th percentile are 0.64 pax/m² and 0.48 pax/m², and the 90th percentiles are 5.69 pax/m² and 5.12 pax/m²).

3. Modeling results

This section presents a general analysis of data collected and factors applied to correct discrepancies between periods. It then shows the specification of MIMIC and the mode choice model, as well as the estimates. Finally, it includes an analysis of marginal rates of substitution and disutility curves.

3.1. Data analysis

We collected 1132 responses, 714 during the first time period and 418 during the second. We paid for targeted Facebook advertisements to reach respondents in Santiago, controlling for their spatial distribution across the city as well as their main sociodemographic characteristics as we gathered the data. Based on the collected data, a general socio-demographic analysis was performed, to analyze the representativity of the database when compared to the latest official data (called Casen, 2017). Regarding gender distribution, we identified an over-representation of women. We obtained a 65% of women's answers but, if we compare to the official distribution of the population, they are 52% (MDS, 2017). Something similar happens for age distribution, the data collected has an over-representation of adults with ages higher than 35 years old in both periods when we compare it with the official distribution (see Fig. 3–1). These differences could be generated by collecting data using social media, and we were bound on the kind of Facebook users. Consequently, we developed a correction factor estimated by the Furness algorithm (Furness, 1965), and then we incorporated this factor into modeling as a weight. In terms of spatial distribution, no corrections were needed as the representativeness of the data (in terms of the 34 municipalities of Santiago) was achieved by default due to how we controlled the Facebook advertisements during the data collection process.

We also analyzed the modal shares and found significant differences

Tables 2–1
Crowding aversion indicators.

N°	Indicator	Name	1st Period Grades			2nd Period Grades		
			1/2	3/4/5	6/7	1/2	3/4/5	6/7
1	I will organize my travel itinerary to avoid crowding in public transport	Organize itinerary	10%	25%	65%	12%	24%	64%
2	I will try to take the first passing bus with no care about crowding level.	Take first bus	42%	37%	20%	47%	35%	18%
3	In order to travel sitting, I am willing to let go buses without available seats.	Let go buses	28%	37%	34%	28%	33%	39%

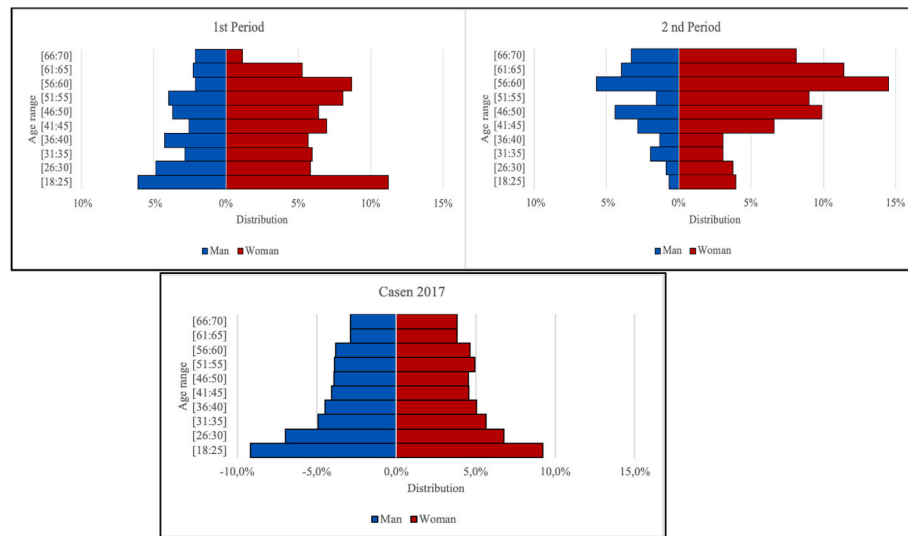


Fig. 3–1. Age and gender distributions.

between both time periods. As the latest Origin-Destination Survey (ODS) is almost a decade old, from 2012, and of course does not consider a pandemic scenario, it is not possible to make a representativity analysis. The modal shares are presented in Fig. 3–2. As a manner to tackle these differences, we decided to incorporate different alternative specific constants for each alternative and time period in the models. This solution allows us to estimate unbiased parameters for the level of service attributes and the latent variable (Ortúzar and Willumsen, 2011).

3.2. Model estimated

To help in the understanding of the model specification, Table 3–1 shows the notation and definition of each variable used in the hybrid discrete choice model.

The hybrid discrete choice model has estimated simultaneously using Pandas Biogeme (Bierlaire and Fétiarison, 2009). Even though the MIMIC model and the mode choice model are estimated simultaneously, they are presented and described separately, to simplify the analysis.

The latent variable of crowding aversion was included as a systematic variation on the travel time of public transport and was estimated through a MIMIC model. To define the measurement equation, where

Tables 3–1

Variable and their definition.

Variables	Definition
<i>Young</i>	Equals to 1 if is younger than 35 years and 0 in another case
<i>Woman</i>	Equals to 1 if is woman and 0 in another case
TC_i	Travel cost (CLP) for alternative i
TT_i	Travel time (minutes) for alternative i
WT_i	Walking time (minutes) for alternative i
$Waiting T_i$	Waiting time (minutes) for alternative i
<i>Density</i>	Public transport density (pax/m ²)
<i>CA</i>	Value of Crowding aversion latent variable
P_1	Equals 1 if it corresponds to first period and 0 if not
P_2	Equals 1 if it corresponds to second period and 0 if not

the latent variable explains the perception indicators, we used the three indicators presented in Table 2–1 (Cronbach's Alpha = 0.82). The structural Equation (7) describes the value of the latent variable was defined using two characteristics from users: women and young. We expected negative parameters for young and positive for women because we hope that women and adult people maximize the crowding aversion. First, women have more harassment and delinquency possibilities on

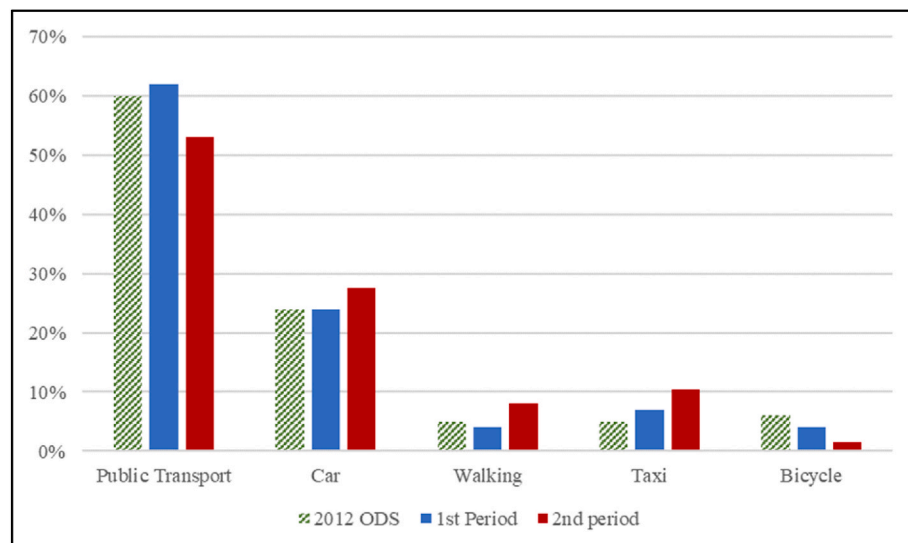


Fig. 3–2. Survey modal shares.

public transport (Gardner et al., 2017; Allen et al., 2019). And second, adults have more health risks when infected with COVID-19.

$$CA = ASC_{CA} + \alpha_{Woman} \bullet Woman + \alpha_{Young} \bullet Young + v_{CA} \quad (7)$$

Table 3–2 presents the estimated parameters for the MIMIC's measurement and structural equations. Regarding the structural equation, α_{Woman} is positive and α_{Young} is negative. As Haywood et al. (2017) conclude, a policy that reduces crowding levels in public transport will have women as principal beneficiaries, which will happen in this case. Regarding the measurement equations, to guarantee the identifiability of the model $\gamma_{CA,3}$ was fixed. This implies that the latent variable interpretation is crowding aversion and not the opposite (crowding sympathy). This way, the other two parameters have the correct sign: on the one hand, $\gamma_{CA,2}$ is negative as a person who prefers to take the first bus independent of crowding level is less crowding averse; on the other hand, $\gamma_{CA,1}$ has a positive sign because a person who chooses to organize his travel and avoid taking a crowded bus means more crowding averse.

With the MIMIC interpretation clear, now we now focus on the mode choice model. The public transport travel time has interactions with the crowding aversion and the passenger density. However, this interaction was found to be statistically significant on first time period. Non-linear effects were tested through Box-Cox transformations (Sakia, 1992). Nonetheless, we found that the Box-Cox power parameter took a value nearly to zero in each model, which suggests a logarithmic effect on the travel time. Thus, as a reference, the utility specification for public transport is given by Equation (8). This is the most complex utility specification of all alternatives, as it is the only mode that has density and crowding aversion.

$$V_{TP} = ASC_{P1PT} \bullet P_1 + ASC_{P2PT} \bullet P_2 + \beta_{TC} \bullet TC_{PT} + \beta_{WT} \bullet WT_{PT} + \beta_{WaitT} \bullet WaitingT_{PT} + \beta_{CA} \bullet (CA \bullet Density) \bullet \log(TT_{PT}) \bullet P_1 + \log(TT_{PT}) \bullet P_2 \quad (8)$$

Table 3–3 summarizes the specification for each utility function. Travel time has a logarithmic transformation for all the alternatives, not only public transport. While for walking time, we used a specific parameter for public transport and another one for walking and bike as active time due to the walking time for public transport is like an access time to alternative. Meanwhile, the active time is the total time spent traveling for walking and bike. Car and bicycle availabilities were obtained directly from the survey, while public transport, taxi and walking are available for everyone.

Table 3–4 presents choice model parameters and the t -values. As the travelers' behavior changes over the two time periods, so does the model specification, therefore Table 3–4 also shows to which time periods do the parameters correspond to. The alternative specific constants on both

Tables 3–3
Structure of each utility function.

Variable	Parameter	Alternative				
		Public Transport	Car	Walking	Taxi	Bicycle
Specific constant	ASC	No	No	No	No	Yes
	Bicycle					
	ASC	No	No	Yes	No	No
	Walking					
	ASC PT	Yes	No	No	No	No
Cost	ASC Taxi	No	No	No	Yes	No
	β_{Cost}	Yes	Yes	No	Yes	Yes
Walking Time	β_{WT}	Yes	No	No	No	No
Active Time	β_{AT}	No	No	Yes	No	Yes
Travel Time	β_{TT}	No	Yes	No	No	No
Waiting Time	β_{WaitT}	Yes	No	No	Yes	No
Crowding Aversion	β_{CA}	Yes	No	No	No	No

Tables 3–4
Discrete choice mode estimates.

Parameter	Estimate	t-value	Period
ASC _{P1} Car	−0.74	−1.87	1st
ASC _{P1} Bicycle	−2.23	−7.30	1st
ASC _{P1} Walking	−2.10	−6.03	1st
ASC _{P1} Taxi	−2.48	−6.06	1st
ASC _{P2} Car	−0.17	−0.95	2nd
ASC _{P2} Bicycle	−2.72	6.31	2nd
ASC _{P2} Walking	1.24	5.03	2nd
ASC _{P2} Taxi	1.02	4.69	2nd
β_{Cost}	−0.065	−2.68	Both
β_{WT}	−0.061	−5.33	Both
β_{TT}	−0.25	−1.94	Both
β_{WaitT}	−0.026	−3.07	Both
β_{CA}	−0.034	−3.24	1st
β_{AT}	−0.029	−7.21	Both

time periods have different magnitudes and significance. This is due to the differences in modal share in both datasets, after all the alternative specific constants allow the model to reproduce the observed modal shares. All time parameters are negative and statistically significant, with B_{WT} 130% larger than B_{WaitT} , which is consistent with the literature as people prefer to spend time waiting still than walking (Raveau et al., 2014). In the same way, B_{CA} , which groups a triple effect (on public transport for the first time period), has a correct negative sign because with higher travel time, crowding aversion or density we have a higher disutility.

One way to analyze the differences between time periods is by comparing the disutility generated by different travel times (which has a non-linear effect). Fig. 3–3 presents the active time necessary to equate the disutility of a travel time from the first period under different crowding levels and travel time from the second period. To simplify the analysis, only adult women are shown. The curve of the second period, without the crowding effect, is similar to the curve of the first period with 4 pax/m² of density. In addition, if we focus on the first period, we can see that a trip 30 min longer under a low crowding level equates to 13,7 min of walking. By contrast, if the trip has a high crowding level, the walking time increases to 41 min. Therefore, women adults are willing to walk longer than the travel time in the bus.

The previous analysis can be extrapolated to the other variables. Fig. 3–4 presents the disutility curves for each alternative with the travel distance on the horizontal axis. The plot has many inflection points where preferences change from one alternative to another. For example, for trips under 4 km, the bicycle is preferred over the car. But, for longer distances, the preference is the opposite. Then, if we compare public transport with the bike, the inflection point is 10 km with a low crowding level. About taxis, we have that over 4 km this alternative is

Tables 3–2
MIMIC estimates.

		Parameter	Estimate	t-value
Structural Equation	Constant	ASC_{CA}	1.44	10.7
	Young	α_{Young}	−0.349	−2.41
	Woman	α_{Woman}	0.363	3.01
	Std. Deviation	v_{CA}	1.06	5.40
Measurement Equation	Organize itinerary	$\gamma_{CA,1}$	0.174	5.26
		$\delta_{CA,1}^1$	0.330	14.0
		$\delta_{CA,1}^2$	0.545	17.4
		$\delta_{CA,1}^3$	0.569	16.0
	Take first bus	$\gamma_{CA,2}$	−0.349	−8.02
		$\delta_{CA,2}^1$	0.308	14.0
		$\delta_{CA,2}^2$	0.491	17.2
		$\delta_{CA,2}^3$	0.499	15.7
	Let go buses	$\gamma_{CA,3}$	1	fixed
		$\delta_{CA,3}^1$	0.297	9.07
		$\delta_{CA,3}^2$	0.654	12.3
		$\delta_{CA,3}^3$	0.762	12.6

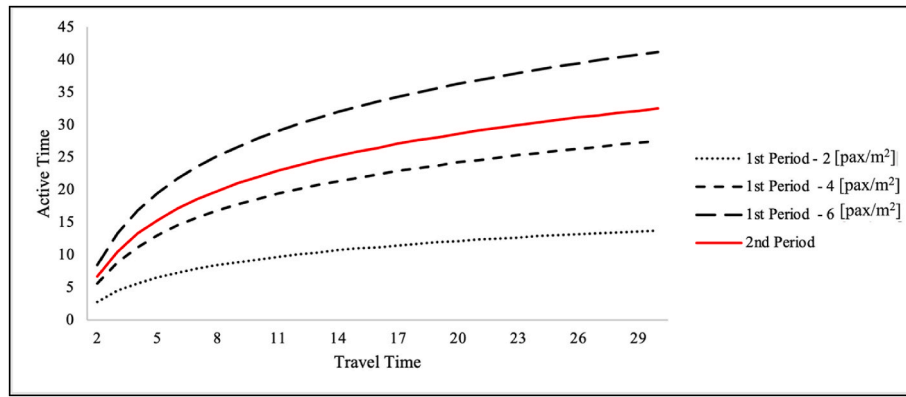


Fig. 3-3. Compensation between travel time and active time.

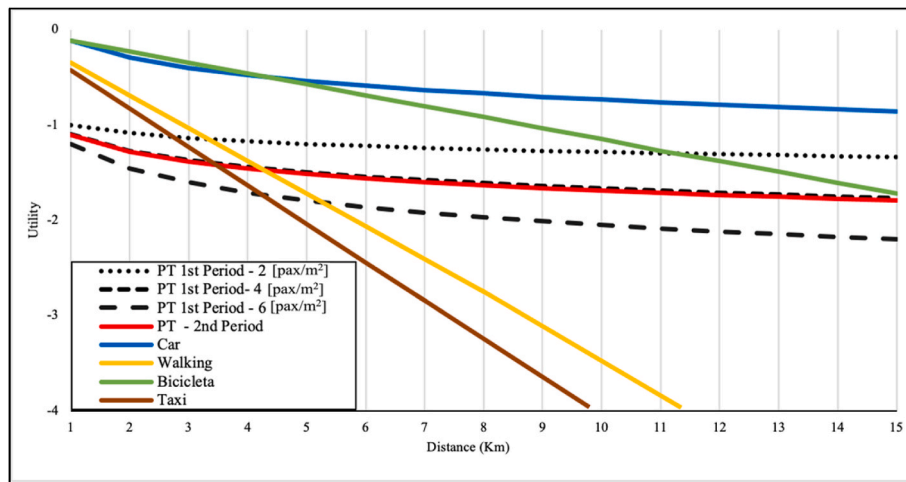


Fig. 3-4. Utilities for each mode and distance.

the worst choice in utility terms. The same happens with walking but has a lower slope.

Marginal rates of substitution (MRS) based on the estimated parameters can also be analyzed. Fig. 3-5 presents the MRS between walking time and travel. As per the previous analysis, we must note that the MRS with travel time on public transport depends on gender, age, density, and crowding aversion, and in this case, we consider also consider adult women. It is seen that adult women are willing to change

2 min on a packed bus for 1 min walk. Meanwhile, when the bus is empty, they will be willing to change 7 min of travel for 1 min walk.

Fig. 3-5 also allows us to analyze MRS evolution on time, and for trips near to 4 min on any crowding condition, the MRS takes values similar to one. In addition, Fig. 3-5 also presents three MRS estimates for Batarce et al. (2015), which allow us to compare our results with a pre-pandemic scenario that used data collected in Santiago. For example, when the crowding level is 6 pax/m², Batarce et al. (2015)

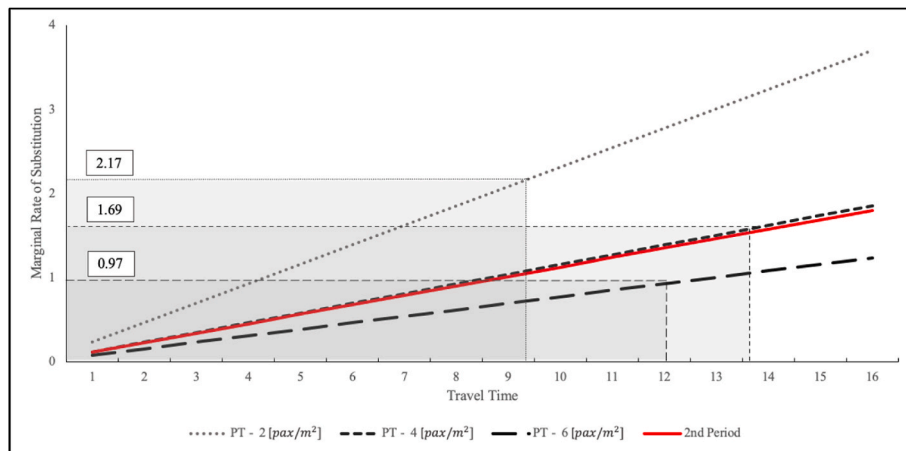


Fig. 3-5. Marginal rate of substitution between waiting time and walking time for public transport.

presented 0.97 MRS for any travel time. By contrast, we have that MRS when travel time is 12 min. Hence, our results show a higher MRS when travel time is higher than 12 min, but we have an MRS less when travel time is less than 12 min. Also, we could say that in a short trip, the effect of passenger density is higher with pandemic than in a pre-pandemic context. Then, if we focus on 2 pax/m² of crowding level, with 9 min, our results have the same MRS as Batarce et al. (2015). Consequently, if we compare our results with the pre-pandemic study, for trips 9 min longer and any crowding level, people in the pandemic prefer to walk than spend the time on public transport. These differences may be due to our investigation taking a different dimension of crowding aversion related to covid infection over dimensions as comfort and security.

4. Conclusions

This study used revealed preference survey data in two time periods to analyze the effect of the COVID-19 pandemic on mode choice, with a special emphasis on the public transport crowding aversion. Since we applied a survey in two different pandemic moments, we can see if crowding perception and mode choice changed between these periods. In this regard, we concluded that from November 2020 to May 2021, the crowding aversion associated with COVID-19 transmission declined. However, this conclusion does not mean that people lost their crowding aversion completely because our latent variable takes just one dimension but not all of them, and we can think that crowding aversion is converging to the initial aversion pre-pandemic (which, instead of sanitary aspects, is based on comfort aspects). These results are relevant as public transport is the alternative that had lost the most modal share participation in the pandemic and is essential to the users' public transport sustainability.

We obtained marginal rates of substitution between travel time and walking time for both periods from the hybrid model. We concluded that for a trip with longer less than 12 min and a high crowding level, the users have a rate of substitution of less than 1, which means that they are willing to change 1 min of traveling for at least 1 min of walk. We can see a point of possible leakage of users in short trips, which put users in the dilemma of walking or taking a bus that could involve possibilities of COVID-19 transmission. Therefore, an action plan is to inform people about the real opportunities of contagion, cleaning and ventilation processes and avoid the migration of public transport users.

From comparing our results with Batarce et al. (2015), we can say that during the first year of the pandemic compared to before it, people on trips of less than 9 min have a greater preference for walking than before the pandemic. People on trips shorter than 9 min have more inclination for walking than spending time inside public transport than pre-pandemic. It must be considered that both studies were conducted in precisely different contexts and with other sample sizes and different contexts using distinct data.

In addition to the rate of substitution analysis, we compared the disutility of each alternative for different travel distances. As a result, we can highlight that in a short trip of fewer than 4 km, the bike is preferred over public transport and the car, and walking is preferred over public transport at least 3.5 km. These results have relevance to the previous analysis because, in a short trip, public transport has multiple competitors, and one of them is the car, which is a pollutant alternative. Thus, if some public transport users migrate to another alternative, they may choose the car. We have a possible loss of users, and we will have more people using the car.

About future research lines, we think that an option to complement our work is doing a new analysis in a future time period. This will help confirm the tendency to decline pandemic effect on mode choice. Also, if the survey is applied again, it is an excellent opportunity to add new questions to address post-pandemic indicators identifying the most important factors for public transport users to feel safe.

Previous and our work show the relevance of crowding aversion on mode choice. Also, we must note that the information of crowding levels

is a crucial input to do an investigation like ours. About information taken from DTPM and Metro, we saw methodology and assumption differences between both. This is relevant because public transport is an integrated system with buses and the metro. Then, we recommend that we must have more compatibility in information generated to not only use data in the inner evaluation of buses and the metro separately but also to do a joint evaluation.

Author statement

The authors confirm that this manuscript has not been previously published and is not currently under consideration by any other journal. Additionally, the authors have read the manuscript, attest to the validity and legitimacy of the data and its interpretation, and agree to its submission to this journal. Finally, to the best of our knowledge, the named authors have no conflict of interest, financial or otherwise.

Authors contributions to the manuscript:

Iglesias, V. Methodology, Data Collection, Validation, Analysis, Investigation, Writing, Raveau, S. Conceptualization, Methodology, Analysis, Writing, Editing, Supervision.

Data availability

The authors do not have permission to share data.

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